

Muhammad Taha

AI Engineer | Computer Vision, OCR, LLM/RAG, Backend Automation
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Summary

AI engineer building production-oriented systems for video understanding, document processing, LLM workflows, and backend automation. I focus on the engineering that makes AI useful after the demo: data quality, evaluation, failure modes, interfaces, deployment, logging, and repeatable operations.

Recent work includes a basketball made-basket clip finder for a US sports client, a document preprocessing pipeline for scanned county records, LLM/RAG systems, guarded data-analysis tools, and full-stack review workflows.

Professional Experience

Basketball Video Analysis System - Freelance AI Engineer

Client project | Mar 2026 - Aug 2026

- Built a computer-vision workflow that scans basketball game footage and exports made-basket highlight clips with timestamped 20-second windows.
- Combined hoop/ball detection, trajectory checks, and a verification stage to reduce false positives while keeping recall high for real makes.
- Delivered a dashboard workflow with queue handling, time-range selection, live logs, Windows launch scripts, and a false-positive review panel for fast client cleanup.
- Received public 5.0 client feedback for speed, flexibility, and quality of work.

Document OCR Preprocessing Pipeline - Client AI Engineer

US records-processing environment | Active client work

- Built a Python/OpenCV preprocessing pipeline for scanned county property records, preserving Year/Month/Date folder structure and multi-page TIFF page order.
- Implemented orientation correction, deskewing, artifact cleanup, denoising, illumination normalization, contrast improvement, and adaptive thresholding.
- Added configuration profiles, resumable execution, failure tracking, runtime logs, worker controls, and Windows operations documentation for approved client infrastructure.

Gohar Textile Mills Pvt Ltd. - AI Engineer Intern

Faisalabad | Jun 2026 - Aug 2026

- Built local AI and data workflows around SAP production logs, sales matrices, and executive reporting needs.
- Developed Python automation pipelines for shift operations, document handling, and data-entry reduction.
- Implemented sales forecasting and backtesting workflows using SAP OData, XGBoost, Prophet, seasonal baselines, MAPE scoring, and Streamlit dashboards.

AWS Cloud Club FAST-NUCES - Co-Lead, Development & Projects

Peshawar | Oct 2025 - Present

- Helped lead cloud-focused student projects, workshops, and mentoring around AWS architecture, deployment, and production habits.
- Guided teams on EC2, Lambda, serverless workflows, CI/CD, and practical project delivery.

Selected Projects

MCP Data Analyst Python, FastAPI, PostgreSQL, MCP, Docker

Built a guarded PostgreSQL analysis service exposing query, statistics, and chart workflows through MCP tools and REST endpoints.

Used SQL AST validation, read-only transactions, statement timeouts, row limits, and query-cost controls around generated SQL.

Rabt / Context Optimization Python, AST parsing, NetworkX, Gemini

Built graph-based code-context tooling that extracts smaller relevant subgraphs for code-aware LLM workflows.

Focused on context reduction, dependency understanding, and inspectable memory/context construction.

QR Payment Verification React, Express, MongoDB, Gemini Vision

Built a receipt-review workflow with OCR extraction, duplicate detection, weighted risk signals, async upload status, and an admin review interface.

SnapTeX Python, Streamlit, Gemini, LaTeX

Built a document conversion workflow that turns images and PDFs into editable LaTeX, with page-level parallel processing and optional PDF compilation.

Education

Bachelor of Science, Computer Science

National University of Computer and Emerging Sciences (FAST-NUCES) | Aug 2023 - Aug 2027 expected

- Relevant coursework: machine learning, data structures and algorithms, advanced databases, compiler construction, operating systems, and computer networks.

Core Skills

AI engineering: computer vision, OCR/document AI, LLM applications, RAG, MCP integrations, prompt engineering, evaluation, failure analysis, and human-in-the-loop workflows.

Backend and data: Python, FastAPI, Flask, REST APIs, PostgreSQL, SQL, MongoDB, Pandas, NumPy, Parquet, data pipelines, validation, and audit-friendly processing.

Computer vision: OpenCV, YOLOv8, EfficientNet, image preprocessing, object detection, video analysis, OCR readiness, perceptual hashing, and risk scoring.

Product delivery: dashboards, client workflows, Docker, Git, Linux, Windows operations, FFmpeg, AWS EC2, Firebase, CI/CD, logging, resumable jobs, and deployment documentation.

Frontend and apps: React, Next.js, TypeScript, JavaScript, Streamlit, Flutter, Dart, Firebase Auth, and Firestore.

How I Work

I do not start with a model choice. I start with the workflow, data quality, constraints, review process, and failure cost. Then I choose the simplest system that can reliably solve the problem: deterministic processing, computer vision, RAG, LLM tooling, or a hybrid.

I care about production details that make clients trust the system: clear setup, repeatable runs, logs, resume behavior, safe defaults, review screens, documented limitations, and maintainable code.

Certifications

Model Context Protocol | AI Fluency: Framework & Foundations | Google Cybersecurity | Claude Code in Action